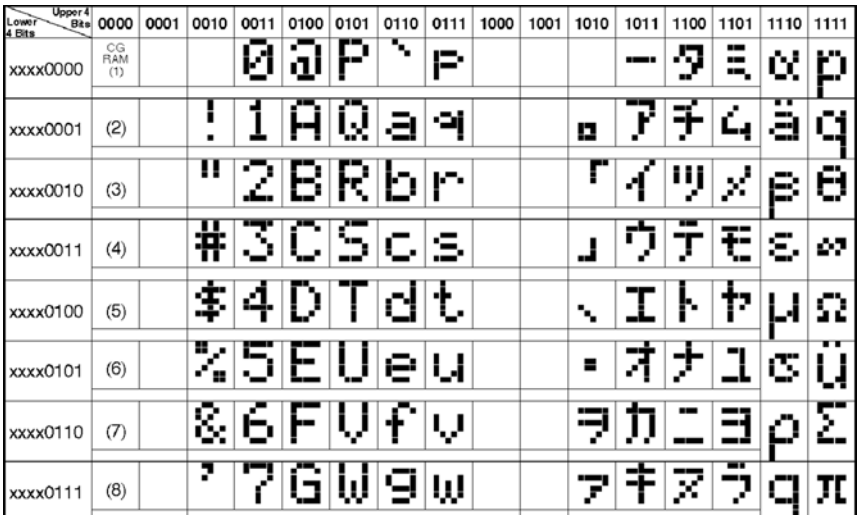


Figure 9-3. *Partial symbol codes from data sheet*

To use a symbol like the Greek omega, Ω, for example, we need to cross reference the first four bits, or 1s and 0s, at the top of the chart with the second four on the left. That will give us the binary number `11110100` that we can then use with the `write()` function to display the symbol on the LCD. Likewise, pi, π, would be the symbol at `11110111`. We can then convert this value to another base if we particularly wanted to and use that as the parameter for the `write()` function as we mentioned earlier.

createChar()

While the LCD has many different symbols built into its driver chip, we might want to create a symbol like a smiley face, music note, or even a fish. Our character display can store eight custom characters numbered 0–7, each occupying a 5×8 grid of pixels. Creating a custom character to display on the LCD is a two-step process. First, we need to define a bitmap array that corresponds to each dot for the character and, second, we need to use the `createChar()` function to send our bitmap to an address in the displays memory. The syntax for this function is as follows:

```
createChar(number, data)
```

The number parameter is the number of the custom character with a range of 0–7. The data is the array name that stores the bitmap image. Before fully explaining our example code, let's look at the following hypothetical statement to make a smiley face:

```
lcd.createChar(0, smiley);
```

This line will create a new character at the number 0 according to the pattern in the array `smiley[]`. One way of creating this array is as follows:

```
byte smiley[] = {B00000,B00000,B01010,B00000,B10001,B01110,B00000,B00000};
```